NASA SPACE APPS 2026 **Climate Lens — Bangladesh**

Official Validation & Limitations Dossier
Out-of-Sample Verification: 2001–2015 Train vs. 2016–2025 Test

Empirical Validation & Limitations Dossier

Rigorous out-of-sample backtesting (640 district-years), error diagnostics, prediction band calibration disclosures, and scientific boundaries.

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Evaluated Scope: 64 Districts · $N = 640$ District-Years

🎥 **Screen Evidence for Judging Video — Section 09 (03:00 – 03:18 · 18s):** This document provides unvarnished empirical proof of held-out model performance and interval calibration.

1 Out-of-Sample Empirical Backtesting Framework

To verify model fidelity and avoid synthetic overconfidence, Climate Lens subjects its projection engine to **strict temporal out-of-sample backtesting** across all 64 districts using authentic daily NASA POWER satellite assimilations (2001–2025):

Training Window (2001–2015 · 15 Years)
Used exclusively to train robust non-parametric **Theil–Sen linear regressions** and compute residual dispersion ($\pm 1.96\sigma$). Models were blinded to post-2015 reality.

Held-Out Test Window (2016–2025 · 10 Years)
Unseen ground truth used to evaluate predictions across 64 districts $\times$ 10 test years = **640 out-of-sample evaluations** for all 4 primary climate indicators.

2 National Test-Period Error & Uncertainty Diagnostics (2016–2025)

MONSOON RAIN (MAE)	ANNUAL RAIN (MAE)	SUMMER PEAK HEAT (MAE)	ROOT SOIL WETNESS (MAE)
639.8 mm	**1,129 mm**	**2.07 °C**	**8.97 %**
RMSE: 778.1 mm	RMSE: 1,300.5 mm	RMSE: 2.53 °C	RMSE: 9.57 %

INDICATOR	OBSERVED BASELINE	MAE	RMSE	95% INTERVAL COVERAGE	EMPIRICAL CALIBRATION DIAGNOSIS
Monsoon Rain (Jun–Sep)	1,531 mm	639.78 mm	778.12 mm	47.3%	Substantial inter-annual monsoon volatility; requires ensemble ribbons.
Annual Precipitation	2,240 mm	1,129.34 mm	1,300.52 mm	20.2%	Non-stationary extremes; poorly bounded by stationary linear models.
Summer Peak Heat (Tmax)	35.8 °C	2.07 °C	2.53 °C	55.0%	Strong thermal trend continuity; tracks long-term planetary warming.
Root Soil Wetness (0–100cm)	62.4%	8.97%	9.57%	21.1%	Heavily decoupled by regional tube-well irrigation & drainage canal shifts.

3 Interval Calibration Disclosure (Scientific Honesty Protocol)

⚖️ **Scientific Integrity Statement: Why Empirical Coverage is 20.2% – 55.0% vs Nominal 95%**

A standard Gaussian model assumes a stationary $\pm 1.96\sigma$ envelope covers 95% of future observations. In Bangladesh's real climate, our held-out test achieves only **20.2% to 55.0%** coverage. **We deliberately report this limitation:** the South Asian summer monsoon is non-stationary and prone to chaotic extremes (e.g. 2022 Sylhet deluge, where coverage plunged to 5.0%). To mitigate this, Climate Lens couples Theil–Sen statistics with **NASA NEX-GDDP-CMIP6 downscaled physical GCM ensembles** to provide dynamic 10th-to-90th percentile uncertainty ribbons.

4 Key Scientific Limitations & Boundaries

1. Spatial Grid Resolution (~50 km)
NASA POWER delivers 0.5° (~55 km) grid points. Small neighboring districts (e.g. Dhaka and Narayanganj) share identical readings; micro-scale urban heat islands cannot be resolved.

2. Exploratory Projections vs Forecasts
Trajectories for 2040/2050 are exploratory scenario projections under IPCC SSP2-4.5/SSP5-8.5. They are **NOT deterministic local weather forecasts** and do not replace BMD advisories.

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Division Matrix & Video Recording Teleprompter
Empirical Dataset: `src/data/validation.json`

5 Division-Level Empirical Validation Matrix (2016–2025 Ground Truth)

Out-of-sample backtesting metrics aggregated across all 8 administrative divisions demonstrate distinct regional hydro-climatic volatility across Bangladesh:

DIVISION	DISTRICTS	MONSOON MAE	MONSOON COV %	HEAT MAE	HEAT COV %	ANNUAL RAIN MAE	RAIN COV %	REGIONAL HYDRO-CLIMATE CHARACTER
Khulna	10	357.67 mm	81.0%	1.60 °C	64.0%	634.49 mm	43.0%	Best calibrated division; stable coastal delta patterns.
Chittagong	11	841.32 mm	48.2%	1.47 °C	65.5%	1,241.80 mm	34.5%	High thermal fidelity; moderate orographic rain variation.
Dhaka	13	598.12 mm	46.9%	2.04 °C	63.1%	1,203.88 mm	8.5%	Dense floodplain; heavy inter-annual rainfall swings.
Rajshahi	8	414.15 mm	45.0%	2.24 °C	57.5%	768.56 mm	10.0%	Barind tract; high summer heat vulnerability.
Barisal	6	1,032.22 mm	45.0%	1.44 °C	58.3%	1,659.10 mm	28.3%	Coastal estuary; cyclonic surge precipitation spikes.
Rangpur	8	429.62 mm	43.8%	3.11 °C	30.0%	793.85 mm	13.8%	Teesta river basin; cold wave/heat extremes increase MAE.
Mymensingh	4	735.98 mm	20.0%	3.37 °C	20.0%	1,536.30 mm	0.0%	Piedmont flood zones; cloudburst rainfall under-covered.
Sylhet	4	1,112.97 mm	5.0%	2.28 °C	52.5%	2,005.91 mm	2.5%	Extreme Meghalaya orographic runoff; severe 2022 flood anomaly.

6 Additional Scientific Limitations & Confounding Nuance

3. Sensor Homogeneity (2001 Start Baseline)
Dataset initiates in 2001 to coincide with TRMM and MODIS satellite assimilations in NASA POWER, preventing artificial pre-2001 sensor transition jumps.

4. Confounding Drivers in Agriculture & Fish
Rice yields and fish stocks are confounded by HYV seed adoption, deep tube well irrigation, and Farakka water diversion. Correlation does not equal isolated climate causality.

7 Judging Video Screen Recording Beat Sheet (03:00 – 03:18 · 18 Seconds)

"I also test the analysis against held-out years. Current prediction ranges still need better calibration. So, future results are exploratory projections, not exact local forecasts."

Duration: 03:00 – 03:18 (18.0 seconds total) Target Pace: 29 English Words · ~98 wpm (Calm, authoritative)

TIMECODE	ON-SCREEN VISUAL ACTION	SPOKEN WORD SYNCHRONIZATION & VOICEOVER CUE
03:00 – 03:04	Display Page 1 Part 1 (Train 2001–2015 vs Test 2016–2025, 640 District-Years).	*"I also test the analysis against held-out years..."*
03:05 – 03:09	Highlight Part 2 Error Table (Monsoon MAE: 639.8 mm, Heat MAE: 2.07 °C).	*"...Current prediction ranges..."*
03:10 – 03:14	Focus on Part 3 Coverage disclosure (Honest 20.2%–55.0% calibration).	*"...still need better calibration..."*
03:15 – 03:18	Show Part 4 Limitations Note ("Exploratory Projections, not exact forecasts").	*"...So, future results are exploratory projections, not exact local forecasts."*